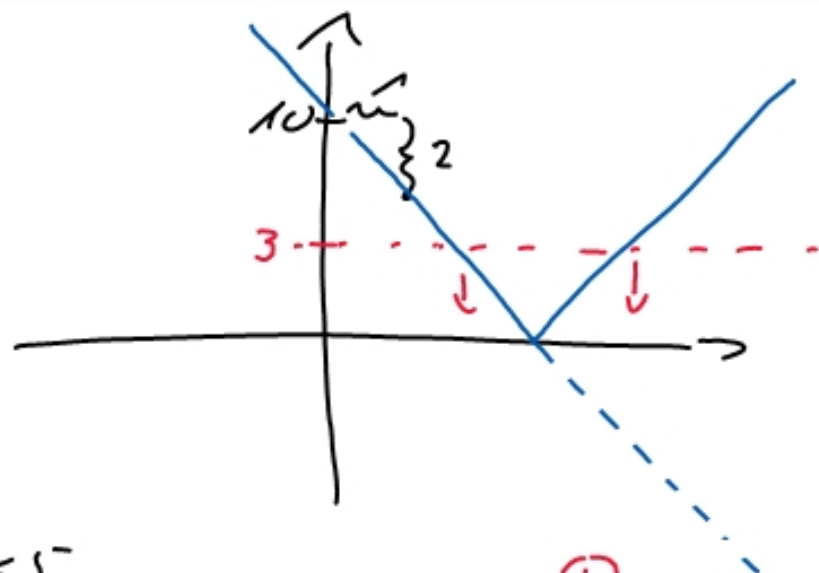


$$\frac{3x}{4x-8} > 2 \quad | \cdot (4x-8) \quad \text{D} = x \in \mathbb{R} \setminus \{2\}$$

F	$x > 2$	$\oplus$	$x < 2$	$\ominus$
P	$3x > 2 \cdot (4x - 8)$		$3x < 2 \cdot (4x - 8)$	
	$3x > 8x - 16 \quad -8x$		$- \quad \quad -$	
	$-5x > -16 \quad : (-5) \quad \text{U}$		$x > 16/5 = 3\frac{1}{5} = 3,2$	
C	$x > 2 \wedge x < 3,2$		$x < 2 \vee x > 3,2$	
P	$x = 3 \cdot \frac{9}{4} > 2 \quad \checkmark$		$x = 0 \quad 0 > 2 \quad \text{falsch}$	
L	$\mathcal{L} = x \in]2; 3,2[$			

$$|10 - 2x| \leq 3$$

$x = 5$



$\bar{I}$	$x \geq 5$	$\ominus$	$x < 5$	$\oplus$
$\mathcal{R}$	$-(10 - 2x) \leq 3$ $-10 + 2x \leq 3$ $2x \leq 13$ $x \leq 6,5$		$10 - 2x \leq 3$ $-2x \leq -7$ $x \geq 3,5$	
$\bar{C}$	$x \geq 5 \wedge x \leq 6,5$		$x < 5 \wedge x \geq 3,5$	
$\mathcal{P}$	$x = 6: 10 - 12 = 2 \leq 3 \checkmark$		$x = 4: 10 - 8 = 2 \leq 3 \checkmark$	
$\mathcal{L}$	$\mathcal{L} = x \in [3,5; 6,5]$			

$$\frac{3}{4-2x} < 4 \quad | \cdot (4-2x) \quad \text{---} \rightarrow x=2$$

$$\text{F} \quad x > 2 \quad \ominus \quad x < 2 \quad \oplus$$

$$\text{R} \quad 3 > 4 \cdot (4-2x)$$

$$3 < 4 \cdot (4-2x)$$

$$3 > 16 - 8x \quad | -16$$

- 11 -

$$-13 > -8x \quad | : (-8) \quad ?$$

$$\frac{13}{8} < x \quad x > \frac{13}{8} = 1\frac{5}{8}$$

$$x < 1\frac{5}{8}$$

$$\text{E} \quad x > 2$$

$$x < 1\frac{5}{8}$$

$$\text{P} \quad x = 3 \quad \frac{3}{-2} < 4 \quad \checkmark$$

$$x = 0 \quad \frac{3}{4} < 4 \quad \checkmark$$

L

$$\mathcal{U} = \{x \in \mathbb{R} \mid x > 2 \vee x < 1\frac{5}{8}\}$$

$$|3x - 6| < 4$$

$$\text{w} : x = 2$$



$$F \quad x \geq 2$$

(+)

$$x < 2$$

(-)

$$R \quad 3x - 6 < 4 \quad | +6$$

$$3x < 10 \quad | :3$$

$$x < 10/3 = 3 \frac{1}{3}$$

$$-(3x - 6) < 4$$

$$-3x + 6 < 4 \quad | -6$$

$$-3x < -2 \quad | \cdot (-1/3)$$

$$x > 2/3$$

(-)

E

$$x \geq 2 \wedge x < 3 \frac{1}{3}$$

$$x > 2/3 \wedge x < 2$$

P

$$x = 3 : |9 - 6| = 3 < 4 \quad \checkmark$$

$$x = 1 : |3 - 6| = |-3| = 3 < 4 \quad \checkmark$$

L

$$x \in] 2/3 ; 3 \frac{1}{3} [$$